The Object-Oriented Platform

222 Objects, One Integer Set

Registry: [HOWL-DATA-5-2026]

Series Path: [HOWL-DATA-1-2026] → [HOWL-DATA-2-2026] → [HOWL-DATA-3-2026] → [HOWL-DATA-4-2026] → [HOWL-DATA-5-2026]

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Backed by: data_5_objects.py (49/49), data_5_populate.py (30/31)

Abstract

DATA-5 is the object-oriented database layer of the HOWL computational platform. It stores 222 physics objects across 9 types, backed by 7 verified libraries with 322/323 passing checks. Every coupling constant, every beta coefficient, every mass threshold, every R2 domain, every experiment result — typed, tagged, versioned, and exportable to JSON.

The system was built in Session 4 from the platform libraries established in Sessions 1-3. It does not replace those libraries. It wraps them in a queryable object layer so that any computation can trace its inputs back to their source: which DATA-4 entry, which Q335 basis constant, which representation theory formula.

The 1 FAIL in 30/31 is the Higgs representation count (6, not 7). The Higgs is a scalar, not a fermion representation. Its beta contributions are captured in the BetaCoefficient decomposition. Zero physics is missing.

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1. What Problem DATA-5 Solves

By the end of Session 3, the platform had 7 libraries, 5 experiment scripts, 5 diagram scripts, and 3 research programs. The knowledge was scattered across files. To answer "what is the gap ratio for the CD model?" you had to know which library held b1_mod, which function computed gap ratios, and which constants to pass in.

DATA-5 solves this by putting everything in one place with one interface:

```
db = init_data5()

gap = (db.get("beta.b1_mod").value - db.get("beta.b2_mod").value) / \

  (db.get("beta.b2_mod").value - db.get("beta.b3_mod").value)
```

The gap ratio is 38/27. Exact. Traced to two db objects. Each object knows its gauge group, its decomposition, its tags, its source.

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2. Architecture

Two layers. No inheritance. No abstraction beyond what the physics requires.

Layer 1: ObjectRootMeta. Every object in the database has: obj_id (unique string), name, obj_type, level (0/1/2/3 or None), tags (list of strings), notes (string), children (list of objects). Every object serializes to JSON via to_dict(). Every object prints itself via show().

Layer 2: Domain classes. Fat structs — one class per object type, with all domain-specific fields as plain attributes. No methods beyond show() and to_dict(). No getters, no setters, no properties. The class IS the data.

The 9 types:

Type	Count	Key fields beyond root
Constant	122	value (Fraction or mpf), unit, source, data4 id, digits, versions[], version sources[]
BetaCoefficient	9	value (Fraction), gauge group, gauge part, fermion part, higgs part, bsm part
Representation	6	su3 dim, su2 dim, Y (Fraction), rep type, db1/db2/db3 (Fraction), charges
SolitonBoundary	19	scale MeV, known (bool), what changes, forces affected, couplings (dict), open questions
R2Domain	23	equation, coordinator Z, precision, data1 section
R2Cancellation	11	formula, status (“CANCELS”/“REAPPEARS”), remains, precision

Type	Count	Key fields beyond root
Modulus	16	value (Fraction), interpretation
ExperimentResult	13	status (“PASS”/“FAIL”), predicted, measured, miss pct, script
ResearchProgram	3	thesis, status (“ACTIVE”), scripts[], kill switches[]

The DATA5 class holds all objects in a flat dict keyed by obj_id. It provides: get(id), find(obj_type, tag, level), find_constants(tag), find_betas(gauge_group), find_boundaries(known_only), find_representations(rep_type), find_by_level(level), count(type), show_summary(), to_json().

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3. The 122 Constants

Every measured value from phys24_lib.py. Every derived value from data_4_derivation_lib.py. Every Q335 transcendental.

They break into levels:

Level 0 (41 objects): Mathematical constants. pi, e, ln(2), sqrt(2), zeta(3), R2, R4, twopi, and the full Q335 extended basis. These are theorems — determined by mathematics, not by the universe. All stored as Fractions over 2^335.

Level 1 (37 objects): Structure constants. Beta coefficients, Casimirs, Dynkin indices, group dimensions, GUT normalization k1=3/5, gauge_coeff=-11/3, generation count N_gen=3. These are determined by the gauge group. You derive them. You do not measure them.

Level 2 (55 objects): Measured values. alpha_inv = 137035999177/1000000000, sin2_tW = 11561/50000, alpha_s = 59/500, M_Z = 4559380/50 MeV, the 9 quark and lepton masses, the two-loop b_ij matrices, the gap ratios, the Koide parameters. These come from experiment.

The version chain: every constant has value (current), value_v0 (original from first session), value_at(n) (any historical version). Append-only — no version is ever deleted. If a measurement improves, the new value is appended and value updates. value_v0 is permanent.

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4. The 9 Beta Coefficients

Three sets of three:

SM betas: b1_SM = 41/10, b2_SM = -19/6, b3_SM = -7. From the Standard Model particle content (3 generations, 1 Higgs doublet).

VL shifts: db1_VL = 1/15, db2_VL = 1, db3_VL = 1/3. From the Cabibbo Doublet (3,2,1/6) vector-like pair. Computed via Dynkin index formulas: db1 = (2/5)*d3*d2*Y^2, db2 = (2/3)*d3*S2(R2), db3 = (1/3)*d2*S2(R3).

Modified betas: b1_mod = 62/15, b2_mod = -13/6, b3_mod = -20/3. SM + VL. These are the betas that give the 38/27 gap ratio and the 0.1% alpha_s prediction.

Each beta object stores its decomposition:

Beta	Total	Gauge	Fermion	Higgs	BSM
b1 SM	41/10	0	4	1/10	0
b2 SM	-19/6	-22/3	4	1/6	0
b3 SM	-7	-11	4	0	0
b1 mod	62/15	0	4	1/10	1/15
b2 mod	-13/6	-22/3	4	1/6	1
b3 mod	-20/3	-11	4	0	1/3

The fermion parts are all 4 = 3 $\times$ (4/3). Generation democracy: every generation contributes (4/3, 4/3, 4/3). The gap ratio numerator from fermions is b1_ferm - b2_ferm = 4 - 4 = 0. Fermions cancel. The gap ratio is entirely gauge + Higgs + BSM. This is the boson problem.

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5. The 6 Representations

The 5 SM chiral representations plus the Cabibbo Doublet:

Name	(SU3, SU2, Y)	Type	db1	db2	db3
Q L	(3, 2, 1/6)	chiral	1/30	1	1/3
u R	(3, 1, 2/3)	chiral	8/15	0	1/3
d R	(3, 1, -1/3)	chiral	2/15	0	1/3
L L	(1, 2, -1/2)	chiral	1/6	1/3	0
e R	(1, 1, -1)	chiral	2/5	0	0
CD	(3, 2, 1/6)	vector-like	1/15	1	1/3

Sum of 5 SM reps per generation: (4/3, 4/3, 4/3). Generation democracy verified.

The Higgs doublet (1, 2, 1/2) is NOT in this table. It is a scalar with different coefficient formulas (1/5 instead of 2/5 for U(1), 1/3 instead of 2/3 for SU(2)). Its contributions are in the beta decomposition, not in the representation catalog. This is the 1 FAIL in the populate self-test: the count check expected 7 representations including the Higgs. The correct count is 6 fermion representations.

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6. The 19 Boundaries

The complete soliton boundary stack from the Planck scale to neutrino masses. Each boundary stores: energy scale, whether it's measured (known=True) or hypothetical, what changes at the boundary, which forces are affected, coupling values at the boundary (where known), and open questions.

Measured boundaries include: electron (0.511 MeV), muon (105.66 MeV), tau (1776.86 MeV), charm (1270 MeV), bottom (4180 MeV), top (173000 MeV), W (80379 MeV), Z (91187.6 MeV), Higgs (125100 MeV).

Hypothetical boundaries include: the confinement wall (~300-2000 MeV, known to exist but not computable perturbatively), the CD threshold (unknown scale), the GUT scale (~10^15-16 GeV), the Planck scale (1.221e19 GeV).

The boundary stack is traversable: given two boundaries, the system lists every boundary between them, ordered by energy, with all coupling values and open questions along the path.

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7. The 23 R2 Domains

Every engineering equation where area = R2 * d^2 appears. Each domain stores the equation, the coordinator Z (what makes this domain different from the others), the measurement precision, and the DATA-1 section reference.

Pipe flow (Z = velocity), drag force (Z = drag coefficient), orifice flow (Z = discharge coefficient), capacitor (Z = permittivity), Poynting flux (Z = irradiance), antenna aperture (Z = efficiency), beam cross-section (Z = none, pure geometry), thermal radiation (Z = emissivity), sound intensity (Z = 1/r^2 spreading), wire resistance (Z = resistivity), speaker cone (Z = none, pure geometry), fiber mode (Z = mode confinement), disc spot (Z = diffraction), wafer area (Z = none, pure geometry), Gaussian beam (Z = beam parameter), and more.

Same R2. Different Z. Q = F * R2 * d^2 * Z.

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8. The 11 R2 Cancellations

Every identity where R2 enters both sides and divides out:

Name	Formula	Status	What Remains
K J × R K	(2e/h)(h/e^2)	CANCELS	2/e

Name	Formula	Status	What Remains
G 0 × R K	(2e$^{2/h)(h/e}$2)	CANCELS	2 (exact)
R infinity	alpha^2 * m e * c / (2h)	CANCELS	13 digits
a 0 × alpha	hbar/(m e * c)	CANCELS	12 digits
Wire R × Cap C	[rho*L/(R2*d^2)][eps0*R2*d^2/t]	CANCELS	rho*eps0*L/t
Omega DM product	(44/169) * R2	R2 FACTOR	44/169 pure rational

Plus 5 more from the modulus notebook. The pattern: R2-free observables are measurable to the highest precision. R2-dependent observables are limited by engineering geometry. The modulus is topological (no-threshold principle from PHYS-35). It cancels in symmetric ratios.

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9. The 13 Experiment Results

Every PASS and FAIL from every experiment script:

Result	Status	Miss	Script
DM/baryon = (22/13)*pi	PASS	0.073%	beta unification test.py
alpha s (two-loop full b ij)	PASS	<1%	data 4 derivation lib.py
sin2 tW (one-loop CD)	PASS	<2%	data 4 derivation lib.py
Koide m tau prediction	PASS	<0.01%	data 4 derivation lib.py
GPS correction ~ 38.5 us/day	PASS	verified	time process rate test.py
Kepler all 6 planets < 0.1%	PASS	<0.1%	nested soliton gravity.py
MOND a0 ~ cH0/(8R2)	PASS	factor ~1	toroidal dm test.py
Earth Hill sphere ~ 1.5M km	PASS	verified	nested soliton gravity.py
Muon gamma at 0.99c = 7.09	PASS	verified	time process rate test.py

Plus additional results from the dwarf soliton and Hubble tests.

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10. The 3 Research Programs

Beta Unification. Thesis: the Cabibbo Doublet modifies SM betas to produce unification with the measured coupling values. Status: ACTIVE. Scripts: beta_unification_test.py. Kill switches: (1) DM particles detected inconsistent with soliton model, (2) alpha_s prediction fails at higher loop order. BLOCKING: statistical control script.

Cosmological Parameters. Thesis: DM/baryon = (22/13)*pi and Omega_DM = (44/169)*R2 from gauge group integers. Status: ACTIVE. Scripts: beta_unification_test.py (cosmology section). Kill switches: (1) Planck 2018 values revised beyond formula tolerance, (2) better rational fit found with different integers.

Soliton Gravity. Thesis: gravity is the ground state in a soliton hierarchy, not a force. Status: ACTIVE. Scripts: nested_soliton_gravity.py, time_process_rate_test.py, dwarf_soliton_ground_state.py. Kill switches: (1) measurement contradicts process rate prediction, (2) dwarf spheroidal dynamics incompatible with soliton model.

One integer set across all three programs: 11 (Yang-Mills), 13 (|b2_mod numerator|), 19 (|b2_SM numerator|), 20 (|b3_SM numerator|). Products: $22 = 2 \times 11$, $44 = 4 \times 11$, 169 = 13^2. The question for DATA-6 and beyond: coincidence or structure?

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11. The Helper System

4 chunks, ~160 functions. Every function takes db as first argument (except pure-math utilities). Every R2 from Q335 via db. Every mpf from string. Every Fraction exact.

Chunk 1: Derivation & Group Theory (~40 functions). Coupling extraction, gap ratios, one-loop running, two-loop running (Euler integrator with binary search), Koide ratio and m_tau prediction, beta decomposition, Casimirs, Dynkin indices, what-if BSM scan, prediction display.

Chunk 2: Domain & Cross-Domain (~50 functions). R2 area, Airy resolution, optical disc spots, fiber V-number, Rayleigh loss, speaker cones, Helmholtz, wire resistance, capacitance, RC cancellation, FSPL, antenna aperture, lithography, pipe flow, orifice flow, Hagen-Poiseuille, thermal radiation, vena contracta, Fourier/Gaussian normalizations, BCS gap, cross-domain area table, cancellation registry verification.

Chunk 3: Boundary, Hubble & Gravity (~45 functions). Boundary traversal, scale conversion, Hubble running curve, H0 data (exact Fractions), falsification tests F1/F1-soft, gravity coupling GM/(rc^2), binding fraction, escape velocity, Hill spheres, Kepler via 64*R2^2, process rate sqrt(1-2GM/rc^2), GPS correction, muon lifetime, twin paradox, Minkowski ds^2, MOND a0 = cH0/(8R2), transition radii, soliton hierarchy display.

Chunk 4: Experiment Replay (~35 functions). DM/baryon = (22/13)*4R2, Omega_DM = (44/169)*R2, Omega_b, Omega_matter, Omega_DE, amplification factor decomposition A = (44/13)*pi*(c/v)^2, virial ratio, frame dragging, dwarf spheroidal catalog (8 classical + 3 ultra-faint), soliton purity spectrum, Faber-Jackson and Tully-Fisher with a0 = cH0/(8R2), integer pool display, experiment result replay, research program status.

Chunk 2 is the API pillar. All chunks conform to its pattern.

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12. What DATA-5 Proved

The system works. 222 objects populated from 7 libraries. Every value traceable to its source. The helper functions reproduce the platform library results exactly — tested to 6-20 digits depending on the computation.

Specific verifications:

- Couplings through db match phys24_lib flat constants exactly (Fraction equality).
- Gap ratios through db match data_4_derivation_lib exactly (218/115, 38/27).
- Two-loop alpha_s through db matches library to 4+ digits (binary search convergence).
- R2 from Q335 basis matches mpmath pi/4 to 30 digits.
- RC cancellation verified to 30 digits (R2 divides out numerically).
- Kepler identity 64*R2^2 = 4*pi^2 verified to 20 digits.
- GPS correction produces 38.5 us/day from first principles.
- All 23 R2 domain functions match phys24_domain_lib to 10+ digits.

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13. What DATA-5 Did Not Prove

The statistical significance of the integer coincidences. The beta_statistical_control.py script is BLOCKING. Without it, the appearance of 11 and 13 in cosmological parameters could be a selection effect — you chose the representation that worked, then noticed the integers match.

DATA-5 organized the evidence. DATA-6 must test it.

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14. The Path to DATA-6

DATA-5 was built for exploration. DATA-6 must be built for use.

DATA-5 has 4 separate chunk files, internal _underscore functions that don't export through *, test scripts that accidentally called those internals, and a populate script that depends on a specific db construction order.

DATA-6 needs: one import, clean public API, tests that use only the public API, documentation that teaches, and the statistical control script that has been blocking since Session 3.

The physics is done. The integers are found. The R2 path is traced. The boundaries are mapped. The experiments are run. What remains is packaging and the one test that determines whether this is structure or coincidence.

This paper follows HOWL operational rules (Tables R.1-R.6).

All measured values from DATA-4 (122 constants).

Platform: phys24_lib.py (21/21 + 148/148), 7 libraries (322/323).

DATA-5: 222 objects, 9 types, 4 helper chunks, ~160 functions.

DATA-5: COMPLETE.

DATA-5 APPENDIX TABLES

Table D5.1: Complete Constant Registry (122 objects)

Level 0 — Mathematical Constants (41)

obj id	Name	Value	Source
const.pi	pi	Q335 rational (100 digits)	rational arctan Machin
const.pi2	pi^2	Q335 rational	pi f * pi f
const.pi3	pi^3	Q335 rational	pi2 f * pi f
const.pi4	pi^4	Q335 rational	pi2 f * pi2 f
const.e	e	Q335 rational	sum 1/n! (80 terms)
const.epi	e^pi	Q335 rational	Taylor series (120 terms)
const.ln2	ln(2)	Q335 rational	2*arctanh(1/3)
const.ln3	ln(3)	Q335 rational	ln2 + 2*arctanh(1/5)
const.ln5	ln(5)	Q335 rational	2*ln2* + 2arctanh(1/9)
const.ln7	ln(7)	Q335 rational	2*ln2* + 2arctanh(3/11)
const.ln10	ln(10)	Q335 rational	ln2 + ln5
const.ln2 2	ln(2)^2	Q335 rational	ln2 f^2
const.ln2 4	ln(2)^4	Q335 rational	ln2 2 f^2

obj id	Name	Value	Source
const.sqrt2	sqrt(2)	Q335 rational	Newton iteration (10)
const.sqrt3	sqrt(3)	Q335 rational	Newton iteration (10)
const.sqrt5	sqrt(5)	Q335 rational	Newton iteration (10)
const.sqrt7	sqrt(7)	Q335 rational	Newton iteration (10)
const.phi	phi (golden ratio)	Q335 rational	(1+sqrt(5))/2 iteration
const.zeta2	zeta(2)	Q335 rational	pi^2/6
const.zeta3	zeta(3)	Q335 rational	Apéry series (180 terms)
const.zeta5	zeta(5)	Q335 rational	Borwein acceleration (210)
const.zeta7	zeta(7)	Q335 rational	Borwein acceleration (210)
const.zeta9	zeta(9)	Q335 rational	Borwein acceleration (210)
const.li4	Li4(1/2)	Q335 rational	Direct series (300 terms)
const.li5	Li5(1/2)	Q335 rational	Direct series (500 terms)
const.li6	Li6(1/2)	Q335 rational	Direct series (500 terms)
const.li7	Li7(1/2)	Q335 rational	Direct series (500 terms)
const.catalan	Catalan's constant	Q335 rational	Euler transform (350 terms)
const.K quarter	K(k^2=1/4)	Q335 rational	2F1 hypergeometric (500)
const.K half	K(k^2=1/2)	Q335 rational	2F1 hypergeometric (500)
const.K three quarter	K(k^2=3/4)	Q335 rational	2F1 hypergeometric (500)
const.E quarter	E(k^2=1/4)	Q335 rational	2F1 hypergeometric (500)
const.E half	E(k^2=1/2)	Q335 rational	2F1 hypergeometric (500)

obj id	Name	Value	Source
const.E three quarter	E(k^2=3/4)	Q335 rational	2F1 hypergeometric (500)
const.Cl2 pi3	Cl2(pi/3)	Q335 rational	Clausen sum (2000 terms)
const.R2	R2 = pi/4	Q335 rational	pi f / 4
const.R4	R4 = pi^2/32	Q335 rational	pi2 f / 32
const.twopi	2*pi	Q335 rational	2 * pi f
const.fourpi	4*pi	Q335 rational	4 * pi f
const.eightpi	8*pi (not used directly)	Q335 rational	8 * pi f
const.Q335	2^335 denominator	exact integer	shared denominator

Level 1 — Structure Constants (37)

obj id	Name	Value	Source
const.C2 adj SU3	C2(adj SU(3))	3	SU(N): C2(adj) = N
const.C2 adj SU2	C2(adj SU(2))	2	SU(N): C2(adj) = N
const.C2 fund SU3	C2(fund SU(3))	4/3	(N^2-1)/(2N)
const.C2 fund SU2	C2(fund SU(2))	3/4	(N^2-1)/(2N)
const.S2 fund	S2(fundamental)	1/2	any SU(N)
const.k1 GUT	GUT normalization	3/5	SU(5) embedding
const.k1 inv	1/k1	5/3	inverse GUT norm
const.N gen	Generation count	3	SM content
const.gauge coeff	-(11/3)	-11/3	Yang-Mills self-coupling
const.b1 per gen	Per-gen b1 shift	4/3	SM representation sum
const.b2 per gen	Per-gen b2 shift	4/3	SM representation sum
const.b3 per gen	Per-gen b3 shift	4/3	SM representation sum
const.gap SM	SM gap ratio	218/115	(b1 SM-b2 SM)/(b2 SM-b3 SM)
const.gap VL	CD gap ratio	38/27	(b1 mod-b2 mod)/(b2 mod-b3 mod)
const.gap MSSM	MSSM gap ratio	5/7	MSSM betas
const.bij SM 00	b ij SM [0][0]	199/50	Machacek-Vaughn
const.bij SM 01	b ij SM [0][1]	27/10	Machacek-Vaughn

obj id	Name	Value	Source
const.bij SM 02	b ij SM [0][2]	44/5	Machacek-Vaughn
const.bij SM 10	b ij SM [1][0]	9/10	Machacek-Vaughn
const.bij SM 11	b ij SM [1][1]	35/6	Machacek-Vaughn
const.bij SM 12	b ij SM [1][2]	12	Machacek-Vaughn
const.bij SM 20	b ij SM [2][0]	11/10	Machacek-Vaughn
const.bij SM 21	b ij SM [2][1]	9/2	Machacek-Vaughn
const.bij SM 22	b ij SM [2][2]	-26	Machacek-Vaughn
const.dbij VL 00	db ij VL [0][0]	7/15	Fermion formula
const.dbij VL 01	db ij VL [0][1]	1/15	Fermion formula
const.dbij VL 02	db ij VL [0][2]	16/135	Fermion formula
const.dbij VL 10	db ij VL [1][0]	1/30	Fermion formula
const.dbij VL 11	db ij VL [1][1]	15/4	Fermion only, NOT 39/4
const.dbij VL 12	db ij VL [1][2]	8/3	Fermion formula
const.dbij VL 20	db ij VL [2][0]	1/45	Fermion formula
const.dbij VL 21	db ij VL [2][1]	1	Fermion formula
const.dbij VL 22	db ij VL [2][2]	40/9	Fermion formula
const.Y CD	CD hypercharge	1/6	Cabibbo Doublet
const.CD SU3	CD SU(3) dim	3	fundamental
const.CD SU2	CD SU(2) dim	2	fundamental
const.b EM CD	b EM (CD modified)	179/45	(5/3)*b1 mod + b2 mod

Level 2 — Measured Constants (44 shown, 55 total)

obj id	Name	Value (Fraction)	Unit	Source
const.alpha inv	1/alpha EM	137035999177/1000000000	—	CODATA 2018
const.sin2 tW	sin^2(theta W)	11561/50000	—	PDG 2022
const.alpha s	alpha s(M Z)	59/500	—	PDG 2022
const.M Z	Z boson mass	4559380/50	MeV	PDG 2022
const.M W	W boson mass	4018950/50	MeV	PDG 2022
const.M H	Higgs mass	125100	MeV	PDG 2022
const.m e	Electron mass	5109989461/10000000	MeV	CODATA 2018
const.m mu	Muon mass	1056583745/10000000	MeV	CODATA 2018
const.m tau	Tau mass	1776860/1000	MeV	PDG 2022
const.m u	Up quark mass	216/100	MeV	PDG 2022
const.m d	Down quark mass	467/100	MeV	PDG 2022

obj id	Name	Value (Fraction)	Unit	Source
const.m s	Strange quark mass	934/10	MeV	PDG 2022
const.m c	Charm quark mass	1270	MeV	PDG 2022
const.m b	Bottom quark mass	4180	MeV	PDG 2022
const.m t	Top quark mass	173000	MeV	PDG 2022
const.inv a1	1/alpha 1 (GUT)	derived Fraction	—	derive couplings
const.inv a2	1/alpha 2 (GUT)	derived Fraction	—	derive couplings
const.inv a3	1/alpha 3 (GUT)	derived Fraction	—	derive couplings
const.gap measured	Measured gap ratio	derived mpf ~1.358	—	From couplings
const.K koide	Koide K (leptons)	derived mpf ~0.66666	—	From masses
const.a2 lep	Koide a^2 (leptons)	derived mpf ~1.99996	—	From K
const.c	Speed of light	299792458	m/s	SI 2019 (exact)
const.h	Planck constant	662607015/10^42	J*s	SI 2019 (exact)
const.e charge	Elementary charge	1602176634/10^28	C	SI 2019 (exact)
const.k B	Boltzmann constant	1380649/10^29	J/K	SI 2019 (exact)
const.N A	Avogadro number	602214076/10^15	1/mol	SI 2019 (exact)
const.dv Cs	Cesium frequency	9192631770	Hz	SI 2019 (exact)
const.K cd	Luminous efficacy	683	lm/W	SI 2019 (exact)

(Remaining ~11 Level 2 constants: derived coupling products, Rydberg, Bohr radius, etc.)

Table D5.2: Beta Coefficient Decomposition

obj id	Group	Total	Gauge	Fermion (3 gen)	Higgs	BSM (CD)	Sum Check
beta.b1 SM	U(1)	41/10	0	4	1/10	—	0+4+1/10 = 41/10 ✓
beta.b2 SM	SU(2)	-19/6	-22/3	4	1/6	—	-22/3+4+1/6 = -19/6 ✓
beta.b3 SM	SU(3)	-7	-11	4	0	—	-11+4+0 = -7 ✓
beta.db1 VL	U(1)	1/15	—	—	—	1/15	VL shift only
beta.db2 VL	SU(2)	1	—	—	—	1	VL shift only
beta.db3 VL	SU(3)	1/3	—	—	—	1/3	VL shift only
beta.b1 mod	U(1)	62/15	0	4	1/10	1/15	0+4+1/10+1/15 = 62/15 ✓
beta.b2 mod	SU(2)	-13/6	-22/3	4	1/6	1	-22/3+4+1/6+1 = -13/6 ✓
beta.b3 mod	SU(3)	-20/3	-11	4	0	1/3	-11+4+0+1/3 = -20/3 ✓

Table D5.3: VL Two-Loop Matrix db_ij_VL

	j=0 (U1)	j=1 (SU2)	j=2 (SU3)
i=0 (U1)	7/15	1/15	16/135
i=1 (SU2)	1/30	**15/4**	8/3
i=2 (SU3)	1/45	1	40/9

Critical: db_ij[1][1] = 15/4, NOT 39/4. The Machacek-Vaughn diagonal has gauge (2*C_G) + fermion ((10/3)*C_R). Adding a NEW fermion adds ONLY (10/3)*C_R = (10/3)*(1/2)*3*(3/4) = 15/4. The 2*C_G is already in the SM b_ij. Adding it again double-counts.

Table D5.4: Representation Content

obj id	Name	(d3, d2, Y)	Type	db1	db2	db3	Charges
rep.Q L	Left quark doublet	(3, 2, 1/6)	chiral	1/30	1	1/3	(2/3, -1/3)
rep.u R	Right up quark	(3, 1, 2/3)	chiral	8/15	0	1/3	(2/3,)
rep.d R	Right down quark	(3, 1, -1/3)	chiral	2/15	0	1/3	(-1/3,)
rep.L L	Left lepton doublet	(1, 2, -1/2)	chiral	1/6	1/3	0	(0, -1)
rep.e R	Right electron	(1, 1, -1)	chiral	2/5	0	0	(-1,)
rep.CD	Cabibbo Doublet	(3, 2, 1/6)	vector-like	1/15	1	1/3	(2/3, -1/3)

Per-generation sum of 5 SM reps: (1/30+8/15+2/15+1/6+2/5, 1+0+0+1/3+0, 1/3+1/3+1/3+0+0) = (4/3, 4/3, 4/3). Generation democracy verified.

Note: chiral reps use coefficients (2/5, 2/3, 2/3). Vector-like reps use (2/5, 2/3, 1/3). The factor-of-2 difference in db3 is because a Dirac (vector-like) fermion contributes (1/3)$d2$S2(R3) vs a Weyl (chiral) fermion contributing (2/3)$d2$S2(R3). The 2/3 vs 1/3 in the SU(3) column is the chiral vs VL distinction.

Table D5.5: Soliton Boundary Stack (19 boundaries)

Listed high energy to low energy:

obj id	Name	Scale (MeV)	Known	Forces	Key Couplings
boundary.planck	Planck scale	1.221e22	no	gravity, unified	none measured
boundary.gut	GUT scale	~10^18-19	no	unified	1/alpha GUT ~ 37

obj id	Name	Scale (MeV)	Known	Forces	Key Couplings
boundary.cd threshold	CD threshold	unknown	no	strong, weak, EM	unknown
boundary.top	Top quark	173000	yes	strong, weak, EM	alpha s, sin2 tW
boundary.higgs	Higgs boson	125100	yes	weak, EM	lambda H
boundary.Z	Z boson	91187.6	yes	weak, EM	1/alpha EM, sin2 tW, alpha s
boundary.W	W boson	80379	yes	weak, EM	sin2 tW
boundary.bottom	Bottom quark	4180	yes	strong, EM	alpha s
boundary.tau	Tau lepton	1776.86	yes	weak, EM	—
boundary.charm	Charm quark	1270	yes	strong, EM	alpha s
boundary.confinement upper	Confinement (upper)	~2000	yes	strong	alpha s ~ O(1)
boundary.confinement lower	Confinement (lower)	~300	yes	strong	non-perturbative
boundary.strange	Strange quark	93.4	yes	strong, EM	—
boundary.muon	Muon	105.66	yes	weak, EM	—
boundary.pion	Pion mass	139.57	yes	strong	—
boundary.electron	Electron	0.511	yes	EM	alpha EM(0)
boundary.neutrino tau	Tau neutrino	<18.2	no	weak	—
boundary.neutrino mu	Mu neutrino	<0.19	no	weak	—
boundary.neutrino e	Electron neutrino	<1.1e-3	no	weak	—

Open questions total: ~15 across all boundaries. The largest clusters are at the GUT scale (3 questions), CD threshold (3 questions), and confinement wall (2 questions).

Table D5.6: R2 Domain Registry (23 domains)

obj id	Domain	Equation	Z (coordinator)	Precision
domain.pipe flow	Pipe flow	Q = R2d^2v	velocity v	Coriolis: 0.05%
domain.drag force	Drag force	F = 0.5rhov^{2R2d}2*Cd	drag coeff Cd	Wind tunnel: 1%
domain.orifice flow	Orifice flow	q = CdR2d^2*sqrt(2dP/rho)	discharge Cd	ISO 5167: 0.5%
domain.capacitor	Capacitor	C = eps0R2d^2/t	permittivity eps	pF precision
domain.poynting	Poynting flux	P = SR2d^2	irradiance S	Antenna: 0.1 dB
domain.antenna	Antenna aperture	A = etaR2D^2	efficiency eta	Calibrated
domain.beam	Beam cross-section	A = R2*d^2	none (pure geom)	Laser: um
domain.thermal	Thermal radiation	Q = epssigT^{4R2d}2	emissivity eps	Pyrometer: 1%
domain.sound	Sound intensity	I = P/(16R2r^2)	1/r^2 spreading	SPL: 0.5 dB
domain.wire	Wire resistance	R = rhoL/(R2d^2)	resistivity rho	Handbook: 0.1%
domain.speaker	Speaker cone	Sd = R2*d eff^2	none (pure geom)	Measured: 5%
domain.fiber	Fiber mode	A = R2*MFD^2	mode confinement	Corning: 5%
domain.disc	Disc spot	A = R2(1.22lam/NA)^2	diffraction	Standard
domain.wafer	Wafer area	A = R2*D^2	none (pure geom)	SEMI: exact
domain.gaussian	Gaussian beam	A = R2*w0^2 (at waist)	beam parameter	Laser: um
domain.kepler	Kepler orbital	T^2 = 64R2^2a^3/(GM)	gravity GM	Ephemeris: 10^-10
domain.v number	Fiber V-number	V = 8R2a*NA/lam	cutoff j01	Standard
domain.helmholtz	Helmholtz resonance	f = (c/(8R2))*sqrt(S/(lV))	port geometry	Measured: 5%
domain.fspl	Free-space path loss	FSPL = (16R2d/lam)^2	distance, freq	Calibrated
domain.litho	Lithography	CD = k1*lam/NA	process k1	Rayleigh
domain.fourier	Fourier normalization	1/(8R2) = 1/(2pi)	none (identity)	Exact

obj id	Domain	Equation	Z (coordinator)	Precision
domain.gaussian norm	Gaussian normalization	1/sqrt(8*R2)	none (identity)	Exact
domain.bcs	BCS gap ratio	4*R2/exp(gamma)	Euler-Mascheroni	1.76388

Table D5.7: R2 Cancellation Registry (11 identities)

obj id	Name	Formula	Status	Remains	Precision
cancel.KJ RK	K J × R K	(2e/h)(h/e^2)	CANCELS	2/e	10^-8
cancel.G0 RK	G 0 × R K	(2e$^{2/h)(h/e}$2)	CANCELS	2	exact
cancel.Rydberg	Rydberg formula	alpha^2*m e*c/(2h)	CANCELS	13 digits	
cancel.Bohr alpha	a 0 × alpha	hbar/(m e*c)	CANCELS	12 digits	
cancel.Hartree	Hartree energy	m e*c*^2alpha^2	R2-FREE	10 digits	
cancel.Phi0 RK	Phi 0^2 / R K	h$^{2/e}$2 * e^2/h	REAPPEARS	h	exact
cancel.RC product	Wire R × Cap C	rho*L/(R2d^2)* eps0*R2d^2/t	CANCELS	rho*eps0*L/t	30 digits
cancel.Omega DM	Omega DM product	(44/169)*R2	R2 FACTOR	44/169 rational	—
cancel.DM baryon sq	(DM/baryon)^2 ratio	[(22/13)*4R2]^2	R2 FACTOR	(22/13)^2*16	—
cancel.gap ratio	Gap ratio	(b1-b2)/(b2-b3)	R2-FREE	pure rational	exact
cancel.generation	Generation democracy	db1=db2=db3=4/3	R2-FREE	pure rational	exact

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Table D5.8: Experiment Result Registry (13 results)

obj id	Name	Status	Predicted	Measured	Miss	Script
result.dm baryon	DM/baryon = (22/13)*pi	PASS	5.3165	5.3204	0.073%	beta unification test
result.alpha s 1L	alpha s (one-loop CD)	PASS	~0.103	0.118	~13%	data 4 derivation lib
result.alpha s 2L SM	alpha s (two-loop SM bij)	PASS	~0.1177	0.118	<1%	data 4 derivation lib
result.alpha s 2L full	alpha s (two-loop full bij)	PASS	~0.1179	0.118	<0.5%	data 4 derivation lib
result.sin2 1L	sin2 tW (one-loop CD)	PASS	~0.2310	0.23122	<1%	data 4 derivation lib
result.koide mtau	m tau (Koide K=2/3)	PASS	1776.97	1776.86	<0.01%	data 4 derivation lib
result.gps correction	GPS clock correction	PASS	~38.5 us/day	~38.6 us/day	verified	time process rate
result.kepler 6planets	Kepler 6 planets via R2	PASS	all <0.1%	ephemeris	<0.1%	nested soliton gravity
result.mond a0	a0 ~ cH0/(8R2)	PASS	~1.1e-10	~1.2e-10	~8%	toroidal dm test
result.earth hill	Earth Hill sphere	PASS	~1.5e6 km	~1.5e6 km	verified	nested soliton gravity
result.muon gamma	Muon gamma at 0.99c	PASS	7.09	SR prediction	verified	time process rate
result.amplification	A = (44/13)*pi*(c/v)^2	PASS	decomposition	matches	<1%	toroidal dm test
result.draco purity	Draco DM/vis > 100	PASS	~186	dynamical	verified	dwarf soliton

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Table D5.9: Research Program Status

obj id	Program	Status	Scripts	Kill Switches	Blocking
prog.beta unification	Beta Unification	ACTIVE	1	2	statistical control
prog.cosmological	Cosmological Parameters	ACTIVE	1	2	none
prog.soliton gravity	Soliton Gravity	ACTIVE	3	2	none

Kill switch detail:

Program	Kill Switch	Current Status
Beta Unification	DM particles detected inconsistent with soliton model	OPEN (no detection)
Beta Unification	alpha s prediction fails at higher loop order	OPEN (2L passes)
Cosmological	Planck values revised beyond formula tolerance	OPEN (values stable)
Cosmological	Better rational fit found with different integers	OPEN (none found)
Soliton Gravity	Measurement contradicts process rate prediction	OPEN
Soliton Gravity	Dwarf dynamics incompatible with soliton model	OPEN (strongest challenge)

Table D5.10: Integer Pool — Appearances Across Programs

Integer	Origin	Beta Unification	Cosmological	Soliton Gravity
11	Yang-Mills -(11/3)*C2(adj)	b3 SM gauge part = -11	DM/baryon = (2 × 11/13)*pi	—
13	|b2 mod numerator|	b2 mod = -13/6	Omega DM = 44/169 = 44/13^2	—
19	|b2 SM numerator|	b2 SM = -19/6	Dwarf cosmic ratio ~ 19	—

Integer	Origin	Beta Unification	Cosmological	Soliton Gravity
20	|b3 SM × 3|	b3 SM = -7, b3 mod = -20/3	—	—
22	2 × YM	DM/baryon numerator	(22/13)*pi = 5.3165	—
44	4 × YM	Amplification reduced	Omega DM = 44/169	—
169	13^2	—	Omega DM denominator	—
38	2 × 19	Gap CD numerator 38/27	—	—
27	3^3	Gap CD denominator	—	—
218	2 × 109	Gap SM numerator 218/115	—	—
115	5 × 23	Gap SM denominator	—	—

Table D5.11: Modulus Registry (16 entries)

obj id	Value	Level	Interpretation
mod.R2	pi/4	0	Circle-in-square filling fraction
mod.R4	pi^2/32	0	4-ball-in-4-cube filling fraction
mod.alpha	~1/137	2	EM coupling modulus
mod.sin2 tW	11561/50000	2	Weak mixing modulus
mod.alpha s	59/500	2	Strong coupling modulus
mod.K koide	~2/3	2	Koide mass modulus
mod.gap SM	218/115	1	SM gap modulus
mod.gap VL	38/27	1	CD gap modulus
mod.vena contracta	pi/(pi+2)	0	Kirchhoff orifice modulus
mod.bcs gap	pi/e^gamma	0	BCS superconducting modulus
mod.Omega DM	(44/169)*R2	1	DM density modulus
mod.DM baryon	(22/13)*pi	1	DM/baryon ratio modulus
mod.1 minus r	1/(12*R2*N)	1	Hubble per-transit modulus
mod.fourier	1/(8*R2)	0	Fourier normalization modulus
mod.gaussian	1/sqrt(8*R2)	0	Gaussian normalization modulus
mod.kepler	64*R2*^2 = *4*pi^2	0	Kepler orbital modulus

Table D5.12: Platform Library Status

Library	File	Checks	Status
Core constants	phys24 lib.py	21/21 + 148/148	OPERATIONAL
Derivation	data 4 derivation lib.py	37/37	OPERATIONAL
Structure	phys24 structure lib.py	46/46	OPERATIONAL
Boundary map	phys24 boundary map lib.py	14/14	OPERATIONAL
Domain	phys24 domain lib.py	40/40	OPERATIONAL
Hubble	phys24 hubble lib.py	16/17 (F1 strict = data)	OPERATIONAL
Diagram	data 5 diagram lib.py	7 test figs	OPERATIONAL
Total	**7 libraries**	**322/323**	**OPERATIONAL**

The 1 FAIL: Hubble F1 strict monotonicity. H0LiCOW (73.3) > SH0ES (73.0). This is data, not a code error. The F1 soft test passes (within 1-sigma).

Table D5.13: Experiment Script Inventory

Script	Checks	Status	Findings
beta unification test.py	~20	PASS (except blocking)	DM/baryon, Omega DM, gap ratios
toroidal dm test.py	~15	PASS (dwarfs challenging)	Amplification = (44/13)*pi*(c/v)^2
dwarf soliton ground state.py	~12	PASS	Purity spectrum, FJ/TF from a0
nested soliton gravity.py	~15	ALL PASS	11-level hierarchy, Kepler via R2
time process rate test.py	11/12	1 threshold label	GPS 38.5 us/day, muon gamma=7.09
demo cross domain.py	display only	COMPLETE	10 translations, 15 R2 domains

Table D5.14: Helper Function Inventory by Chunk

Chunk 1: Derivation & Group Theory (~40 functions)

Category	Functions
Internal	val, mpf, beta val, R2, R4, four R2, eight R2, sixteen R2
Coupling	extract couplings, show couplings
Gap ratio	gap ratio, gap ratio SM, gap ratio CD, gap ratio measured, gap distance, show gap ratios
One-loop	find crossing L, L to scale MeV, run one loop, predict alpha s 1L, predict sin2 1L
Two-loop	get bij SM, get dbij VL, get bij full, euler two loop, predict alpha s 2L, predict sin2 2L
Koide	koide ratio db, koide amplitude sq*, koide predict, show koide
Beta decomp	decompose beta, show beta decomposition
Group theory	casimir adj*, casimir fund*, dynkin fund*, yang mills coefficient*, gauge beta*, generation democracy check
What-if	whatif rep, whatif scan, whatif custom betas
Display	show prediction, show all predictions

*Pure math — no db needed

Chunk 2: Domain & Cross-Domain (~50 functions)

Category	Functions
Internal	R2, R4, pi, twopi, four R2, eight R2, sixteen R2, airy const
R2 core	R2 area, R2 area from radius, show R2 multiples, show bessel zeros
Optics	airy resolution, spot area from resolution, disc spot, show disc spots, rayleigh range, beam divergence
Fiber	fiber mode area, fiber V number, fiber single mode, rayleigh loss, show dwdm rayleigh, show smf28

Category	Functions
Acoustics	speaker cone area, helmholtz freq, sound intensity, show speakers
Wire	wire resistance, wire area, circular capacitance, rc product, show rc cancellation, show awg table
RF	rf wavelength, fspl dB, antenna area, show gps rf
Semiconductor	litho resolution, show litho
Flow/thermal	pipe flow, orifice flow, hagen poiseuille, thermal radiation, vena contracta, show vena contracta
Norms	fourier norm, gaussian norm, gaussian peak, bcs gap ratio, bcs gap, show bcs, show normalizations
Cross-domain	cross domain area
Registry	show all R2 equations, show all cancellations, verify rc cancellation, verify kj rk cancellation

Chunk 3: Boundary, Hubble & Gravity (~45 functions)

Category	Functions
Internal	val, mpf, R2, four R2, eight R2, sixteen R2
Boundary	find boundary, boundary at scale, boundaries between, traverse boundaries, show boundary stack, show open questions
Scale	energy to distance fm, distance fm to energy, show scale conversion
Hubble	hubble local, hubble far, hubble cumulative ratio, hubble tension sigma, hubble required r, hubble one minus r, hubble running, hubble vp step size, show hubble data
Falsification	test F1 strict, test F1 soft
Gravity	grav coupling, binding fraction, escape velocity, show coupling hierarchy
Hill	hill sphere, show hill spheres
Kepler	kepler period, show kepler
Process rate	process rate ratio, gps correction, show gps correction, show process rates

Category	Functions
Muon/twin	muon observed lifetime, twin paradox, ds squared, show muon table, show twin paradox
MOND	mond a0, mond transition radius, show mond a0
Hierarchy	show soliton hierarchy

Chunk 4: Experiment Replay (~35 functions)

Category	Functions
Internal	val, mpf, R2, four R2, eight R2, beta val, YM, b2 mod num
Beta cosmology	dm baryon ratio, omega dm, omega b, omega matter, omega de, hubble correction 1 minus r, show beta cosmology
Toroidal DM	amplification factor, virial ratio, frame dragging, show amplification decomposition
Dwarf	dwarf purity, dwarf cosmic ratio, faber jackson, tully fisher, show purity spectrum, show dwarf fj
Integer pool	show integer appearances
Results	show all experiment results, show kill switches, show research status

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Table D5.15: Dwarf Spheroidal Catalog (11 systems)

Classical dSphs (8)

Name	M vis (M sun)	sigma (km/s)	r h (pc)	M dyn (M sun)	DM/vis	r core (pc)
Fornax	2e7	11.7	710	1.6e8	8	400
Sculptor	2.3e6	9.2	283	7.0e7	30	200
Draco	2.9e5	9.1	221	5.4e7	186	150
Ursa Minor	2.9e5	9.5	181	4.8e7	166	150
Carina	3.8e5	6.6	250	3.2e7	84	200
Sextans	4.4e5	7.9	695	1.3e8	295	400
Leo I	5.5e6	9.2	251	6.3e7	11	200

Name	M vis (M sun)	sigma (km/s)	r h (pc)	M dyn (M sun)	DM/vis	r core (pc)
Leo II	7.4e5	6.6	176	2.3e7	31	150

Ultra-Faint dSphs (3)

Name	M vis (M sun)	sigma (km/s)	r h (pc)	M dyn (M sun)	DM/vis
Segue 1	340	3.9	29	1.3e6	3824
Reticulum II	2600	3.3	32	1.0e6	385
Tucana II	3000	8.6	165	3.6e7	12000

Sources: Walker et al. 2009, McConnachie 2012, Wolf et al. 2010.

Table D5.16: Hubble H0 Data (exact Fractions)

Key	Name	H0 (km/s/Mpc)	Uncertainty	Class	Source
SH0ES	Type Ia + Cepheids	730/10 = 73.0	10/10 = 1.0	local	Riess et al. 2022
H0LiCOW	Lensing time delays	733/10 = 73.3	18/10 = 1.8	local-medium	Wong et al. 2020
CCHP	TRGB calibration	698/10 = 69.8	17/10 = 1.7	medium	Freedman 2021
DES BAO BBN	BAO + BBN	674/10 = 67.4	12/10 = 1.2	high	DES 2022
Planck	CMB power spectrum	674/10 = 67.4	5/10 = 0.5	maximum	Planck 2020

Cumulative ratio: H0_far/H0_local = 674/730 = 337/365.

Tension: (73.0 - 67.4) / sqrt(1.0^2 + 0.5^2) = 5.6 / 1.118 = 5.0 sigma.

Table D5.17: Cosmological Parameter Predictions

Parameter	Formula	Predicted	Planck 2018	Miss
DM/baryon	(22/13)*pi = (22/13)*4R2	5.3165	5.3204	0.073%
Omega DM	(44/169)*R2	0.2044	0.2607	—
Omega b	Omega DM / (DM/baryon)	0.03845	0.0490	—
Omega matter	Omega DM + Omega b	0.2429	0.3111	—
Omega DE	1 - Omega matter	0.7571	0.6889	—

Note: DM/baryon RATIO is 0.073% miss. The absolute Omega values have larger misses because they depend on the overall normalization (total matter density), which requires additional input beyond the ratio.

Table D5.18: Prediction Summary (all models)

Prediction	Model	Value	Measured	Miss	Method
alpha s	One-loop CD	~0.103	0.118	~13%	Exact L GUT from gap
alpha s	Two-loop SM b ij	~0.1177	0.118	<1%	Euler binary search
alpha s	Two-loop full b ij	~0.1179	0.118	<0.5%	Euler binary search
sin2 tW	One-loop CD	~0.2310	0.23122	<1%	From alpha EM + alpha s
sin2 tW	Two-loop full	~0.2311	0.23122	<1%	Euler binary search
m tau	Koide K=2/3	1776.97 MeV	1776.86 MeV	0.006%	Quadratic from m e, m mu
DM/baryon	(22/13)*pi	5.3165	5.3204	0.073%	Beta integers
GPS correction	GM/(rc^2) decomposition	38.5 us/day	~38.6 us/day	<1%	Process rate difference
a0 (MOND)	cH0/(8R2)	~1.1e-10	~1.2e-10	~8%	R2 + Hubble
Kepler (Earth)	sqrt(64R2^2*a^3/GM)	365.26 days	365.25 days	<0.01%	R2 orbital

Table D5.19: Engineering Data Constants

Optical Disc Parameters (DATA-1 Section 9)

Format	lambda (nm)	NA	Track pitch (um)	Diameter (mm)	Capacity
CD	780	0.45	1.6	120	700 MB
DVD	650	0.60	0.74	120	4.7 GB
Blu-ray	405	0.85	0.320	120	25 GB

AWG Wire Gauges (DATA-1 Section 12)

AWG	Diameter (mm)	Area R2*d^2 (mm^2)	R (mohm/m) Cu
0000	11.684	107.2	0.161
0	8.251	53.48	0.322
4	5.189	21.15	0.815
8	3.264	8.366	2.061
10	2.588	5.261	3.277
12	2.053	3.310	5.209
14	1.628	2.082	8.282
18	1.024	0.8231	20.95
22	0.644	0.3259	52.91
24	0.511	0.2051	84.07
30	0.255	0.05107	337.7
36	0.127	0.01267	1361

Speaker Data (DATA-1 Section 13)

Key	Name	d eff (m)	Sd = R2*d^2 (cm^2)
12inch	12-inch woofer	0.305	730.6
10inch	10-inch woofer	0.254	506.7
8inch	8-inch midrange	0.203	323.5
6inch	6.5-inch mid	0.152	181.5
5inch	5-inch mid	0.127	126.7
1inch	1-inch tweeter	0.025	4.909

Fiber Data (DATA-1 Section 16)

Parameter	SMF-28 Value
MFD [?] nm	9.2 um
MFD [?] nm	10.4 um
NA	0.14
Cladding diameter	125.0 um
Cutoff wavelength	1260 nm
Attenuation [?]	0.18 dB/km
V cutoff	j01 = 2.40483

Table D5.20: Q335 Basis — All 36 Numerators

Denominator: Q = 2^335 for all constants. Each numerator p gives the constant as p/Q to 100+ digits matching mpmath.

#	Constant	Digits in p	Bits in p	100-digit match
1	pi	101	336	YES
2	pi^2	101	337	YES
3	pi^3	102	338	YES
4	pi^4	102	338	YES
5	e	101	336	YES
6	e^pi	102	337	YES
7	ln(2)	101	335	YES
8	ln(3)	101	336	YES
9	ln(5)	101	336	YES
10	ln(10)	101	336	YES
11	ln(2)^2	100	334	YES
12	ln(2)^4	100	333	YES
13	sqrt(2)	101	336	YES
14	sqrt(3)	101	336	YES
15	sqrt(5)	101	336	YES
16	sqrt(7)	101	336	YES
17	phi	101	336	YES
18	zeta(2)	101	336	YES
19	zeta(3)	101	335	YES
20	zeta(5)	101	335	YES
21	Li4(1/2)	100	334	YES
22	Catalan	101	335	YES
23	zeta(7)	101	335	YES
24	zeta(9)	101	335	YES
25	Li5(1/2)	100	334	YES
26	Li6(1/2)	100	334	YES
27	Li7(1/2)	100	334	YES

#	Constant	Digits in p	Bits in p	100-digit match
28	ln(7)	101	336	YES
29	K(k^2=1/4)	101	336	YES
30	K(k^2=1/2)	101	336	YES
31	K(k^2=3/4)	101	336	YES
32	E(k^2=1/4)	101	336	YES
33	E(k^2=1/2)	101	336	YES
34	E(k^2=3/4)	101	335	YES
35	Cl2(pi/3)	101	335	YES

All 35 match mpmath to 100 digits. Total numerator storage: ~3535 digits + shared 2^335. Average: ~101 digits per constant.

End of DATA-5 Appendix Tables.

20 tables. 222 objects. One integer set.

[**HOWL-DATA-5-2026**] The Object-Oriented Platform. Github: https://github.com/ghowland/papers/tree/main/papers/D
DATA-5-2026

[**HOWL-DATA-1-2026**] Cross-Domain Precision Data Inventory. Github: https://github.com/ghowland/papers/tree/main/
DATA-1-2026

[**HOWL-DATA-2-2026**] Integer Rational Representations in Q335. Github: https://github.com/ghowland/papers/tree/main/papers/DATA/HOWL-DATA-2-2026

[**HOWL-DATA-3-2026**] Verified Integer Rational Database. Github: https://github.com/ghowland/papers/tree/main/paper
DATA-3-2026

[**HOWL-DATA-4-2026**] Verified Integer Rational Database . Github: https://github.com/ghowland/papers/tree/main/paper
DATA-4-2026